

Judhajeet Basu

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RESEARCH INTERESTS

My research centers on **time-domain astronomy** with an emphasis on multiwavelength observations of **novae** and the role of recurrent novae in the **Type Ia supernovae** progenitor problem. I combine UV and X-ray observations with ground-based optical imaging and spectroscopy to probe the physics of transients. I also contribute to survey science and pipeline development for transient detection, and maintain active interests in fast (**FXTs**, **FBOTs**, **GRBs**, **CVs**) and slow transients (**LRNe**, **CL-AGN**).

RESEARCH EXPERIENCES

Postdoctoral Researcher

Indian Institute of Astrophysics

Aug 2025 – present

Bengaluru, India

- Time-domain survey science: novae, CVs, LRNe, supernovae, FXTs, FBOTs, changing-look AGN.

Doctoral Researcher (PhD)

Indian Institute of Astrophysics

Sep 2020 – Aug 2025

Bengaluru, India

- Thesis:* Multiwavelength study of novae in M31. *Supervisor:* Dr. Sudhanshu Barway
- Member of the GROWTH-India Telescope collaboration and the forthcoming Daksha X-ray mission team.

MSc Research Project

Indian Institute of Technology Kharagpur

Jul 2019 – Jun 2020

Kharagpur, India

- Quasinormal modes in curved spacetimes.

Supervisor: Prof. Sayan Kar

Research Intern

Nicolaus Copernicus University

May – Jul 2019

Toruń, Poland

- Cosmic topology and cosmic N-body simulations.

Supervisor: Prof. Boudewijn F. Roukema

Research Intern

Indian Institute of Astrophysics

May – Jun 2018

Bengaluru, India

- Solar chromospheric rotation rates from Ca II-K plage data.

Supervisor: Prof. Dipankar Banerjee

Research Intern

Aryabhatta Research Institute of Observational Sciences

May – Jul 2017

Nainital, India

- Spectral energy distributions of galaxies.

Supervisor: Dr. Amitesh Omar

EDUCATION

Doctor of Philosophy

Indian Institute of Astrophysics

Sep 2020 – Aug 2025

Bengaluru, India

Thesis: Multiwavelength Study of Novae in M31

5-year Integrated MSc in Physics

Indian Institute of Technology Kharagpur

Jul 2015 – Jun 2020

Kharagpur, India

Major: Physics | Micro-specialisation: Simulation Methods and Applications

CGPA: **9.00 / 10**

Senior School Certificate (CBSE)

Central Model School Barrackpore

Apr 2013 – Apr 2015

Kolkata, India

Physics, Mathematics, Chemistry, Biology

94%

OBSERVATIONAL & TECHNICAL EXPERTISE

Facilities (PI)	HCT (114 hr), GIT (M31 daily-cadence survey), AstroSat/UVIT (59 ks), AstroSat/SXT (140 ks)
Facilities (CoI)	HCT (194 hr), AstroSat/SXT (58 ks)
Archives	Swift (UVOT, XRT), ZTF, HST, Keck, P200, CFHT
Data Reduction	IRAF, PyRAF, CCDLAB, HEASoft (XSELECT, XIMAGE)
Spectral Modelling	XSPEC, CLOUDY, PHOENIX
Computing	Python, C, MATLAB, Linux

AWARDS, GRANTS & FELLOWSHIPS

- ASI Zubin Kembhavi Award (as part of the GROWTH-India Team) 2026
 - ANRF International Travel Support (ITS) 2025
 - ASI AstroSat Prize — Honourable Mention (2024) 2025
 - IIA In-house Symposium Best Poster — Runner-up 2024
 - Institute Junior & Senior Research Fellowship (JRF/SRF), IIA 2020–2025
 - Foundation for Polish Science TEAM Project Grant & Travel Allowance 2019
 - Toruń Astrophysics/Physics Summer Programme (TAPS) Allowance 2019
 - IIA Summer Programme Fellowship (IIA SPF) 2018
 - Indian Academy of Sciences Summer Research Fellowship (IASc SRFP) 2017
 - Kishore Vaigyanik Protsahan Yojana (KVPY) National Science Fellowship 2015–2020
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REFEREED PUBLICATIONS

1. **Survival of the accretion disk in LMC Recurrent Nova 1968-12a: UV–X-ray case study of the 2024 eruption**
Basu J., Anupama G. C., Ness J.-U., et al. *ApJ*, 994, 229 (2025)
2. **Challenging Classical Paradigms: Recurrent Nova M31N 2017-01e, a Be/WD System in M31?**
Chamoli S., Basu J., Barway S., et al. *ApJ*, 991, 174 (2025)
3. **Discovery and Detailed Study of the M31 Classical Nova AT 2023tkw: Evidence for Internal Shocks**
Basu J., Kumar R., Anupama G. C., et al. *ApJ*, 980, 129 (2025)
4. **Exploring the Archives: A Search for Novae in UVIT Snapshots of M31**
Basu J., Krishnendu S., Barway S., et al. *ApJ*, 971, 8 (2024)
5. **Multiwavelength Observations of Multiple Eruptions of the Recurrent Nova M31N 2008-12a**
Basu J., Pavana M., Anupama G. C., et al. *ApJ*, 966, 44 (2024)
6. **Searching for GW Optical Counterparts with ZTF: Summary of O4a**
Ahumada T., Anand S., Coughlin M. W., ... Basu J., et al. *PASP*, 136, 114201 (2024)
7. **Science with the Daksha High-Energy Transients Mission**
Bhalerao V., Sawant D., Pai A., ... Basu J., et al. *Exp. Astron.*, 57, 23 (2024)
8. **Differential Rotation of the Solar Chromosphere: A Century-long Perspective from Kodaikanal Ca II K Data**
Mishra D. K., Routh S., Jha B. K., Basu J., et al. *ApJ*, 961, 40 (2024)
9. **Far-UV to Near-IR Observations of SN 2023ixf: A High-Energy Explosion in Complex CSM**
Teja R. S., Singh A., Basu J., et al. *ApJL*, 954, L12 (2023)
10. **India's First Robotic Eye for Time-Domain Astrophysics: The GROWTH-India Telescope**
Kumar H., Bhalerao V., Anupama G. C., ... Basu J., et al. *AJ*, 164, 90 (2022)

Full list including non-refereed publications:  ORCID |  NASA ADS Library

TALKS & POSTERS

Invited Seminars & Colloquia

- Accretion, Recurrence, and Shocks: Novae in Andromeda (M31)* Apr 2026
Seminar, IUCAA, Pune (India)
- Chasing Nova Explosions: From Surveys to Recurrence to Shockwaves* Sep 2025
Seminar, NAOC, Beijing (China)
- Chasing Nova Explosions: From Surveys to Recurrence to Shockwaves* Sep 2025
Seminar, KIAA, Peking University (China)
- Multi-wavelength Study of an Extra-Galactic Recurrent Nova in M31* Nov 2022
Seminar, IIT Bombay (India)

Conference Talks

- Disks Don't Die: AstroSat's UV-X-ray Insights into Extragalactic Novae* Jan–Feb 2026
A Decade of AstroSat Observations, URSC, Bangalore (India)
- Chasing Nova Explosions: From Surveys to Recurrence to Shockwaves* Jun 2025
EAS Annual Meeting 2025, SS43a: Novae, UCC NUI Cork (Ireland)
- UV and X-Ray Study of Novae in M31 and LMC with AstroSat and Swift* Apr 2025
IIA In-house Symposium, IIA Bangalore (India)
- "Shock"-ing Results from M31: Analysis of the Slow Classical Nova AT2023tkw* Feb 2025
ASI 43rd Meeting, NIT Rourkela (India)
- Exploring the Archives: A Search for Novae in UVIT Snapshots of M31* Oct 2024
NUVA e-Meeting: The Impact of UV Surveys in Astronomy, Online
- Multiwavelength Observations and Spectral Modelling of M31N 2008-12a* Apr 2023
RETICO-V, KSO Kodaikanal (India)

Posters & Flash Talks

- Chasing Nova Explosions: From Surveys to Recurrence to Shockwaves* Aug 2025
Binary Stars in a New Era, Lijiang (China)
- Detailed UV-X-Ray Study of Novae in M31 and LMC with AstroSat and Swift* Mar 2025
Transients from Space, STScI Baltimore (USA)
- Multi-wavelength Observations of Multiple Eruptions of M31N 2008-12a* Mar 2024
IIA In-house Symposium, Bangalore (India)
- M31 Nova Survey using GIT and UVIT with a Focus on M31N 2008-12a* Jan–Feb 2024
ASI 42nd Meeting, IISc Bangalore (India)
- Multi-wavelength Study of an Extra-Galactic Recurrent Nova in M31* Nov 2022
Young Astronomers' Meet (YAM), ARIES Nainital (India)
- AstroSat UVIT and SXT Observations of Novae in M31* Sep 2022
Celebrating Seven Years of AstroSat, ISRO HQ Bangalore (India)

SCHOOLS & WORKSHOPS

- ICESUN Summer School 2025: Binary Stars and Compact Objects, Yunnan Obs. (China) Aug 2025
- AI/ML Applications in Astronomy: Generative AI & Agentic Coding , IIA Bangalore (India) May 2025
- ZTF Summer School 2023: Transients (online), U. Minnesota (USA) Jul 2023
- First UVEX Community Workshop (online), Pasadena (USA) Mar 2023
- IFCAM III: SKA/MSE and Multiwavelength Synergies, IIA Bangalore (India) Jan 2023
- ZTF Summer School 2022: Multi-messenger Astrophysics (online), U. Minnesota (USA) Jul 2022
- Gaia Symposium: DR3 and Beyond, IIA Bangalore (India) Jul 2022

Astronomical Data Analysis using Python, IUCAA Pune (India)	Nov–Dec 2021
Summer School in Astrophysics, KSO Kodaikanal (India)	May 2018
National Science (Vijyoshi) Camp, IISER Kolkata	Dec 2014 & Dec 2015

TEACHING & MENTORSHIP

- **Research Mentor**, Indian Institute of Astrophysics 2022–2025
Supervised 4 undergraduates and 2 graduate students on nova and transient projects.
- **Volunteer Teacher**, Primary school students, Bengaluru 2021–2023
Science tutoring during the COVID-19 pandemic.
- **Peer Mentor**, IIT Kharagpur 2018–2020
Mentored 1st and 2nd year undergraduate physics students.
- **Instructor**, National Service Scheme, Govt. of India 2015–2017
Science teaching at a local high school, Kharagpur.

OUTREACH & SCIENCE COMMUNICATION

- **Poster Presenter & Volunteer**, National Science Day, IIA Bengaluru. 2022–2026
Poster: “Time-Domain Astronomy: Transients”
- **Organizer**, public stargazing events, IIA Bengaluru. 2022–2025
- **Science Writer**, article for [DOOT e-magazine](#), IIA. 2024-2026
- **Volunteer**, National Space Science Symposium Exhibition, Kolkata. Dec 2022
- **Organizer**, National Students’ Space Challenge, IIT Kharagpur. 2015–2017
- **Contributor**, space science quizzes & puzzles, *Moonwalk* magazine, IIT Kharagpur. 2015–2017
- **Organizer**, stargazing events, IIT Kharagpur Astronomy Club. 2015–2017

WORK IN MEDIA

- **Novae in UVIT M31 Archive**
[Nature India](#) | [DST, Govt. of India](#) | [Deccan Herald](#) | [The Hindu](#)
- **M31N 2008-12a recurrent nova study**
[AAS Nova](#)